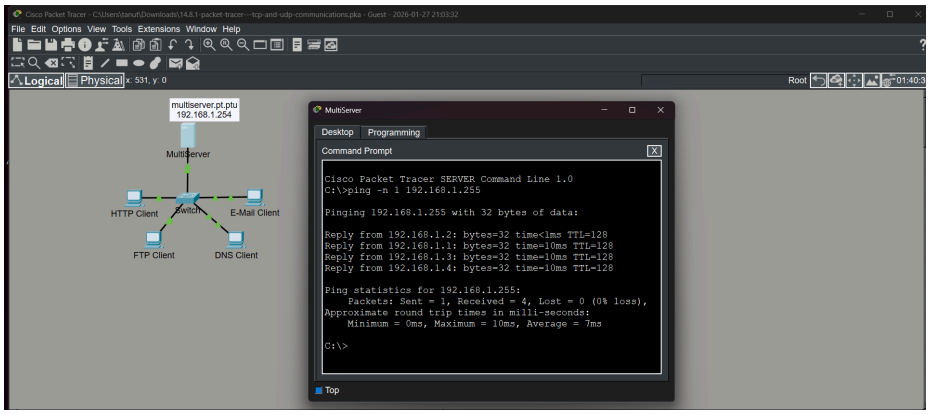
673380041-0 ธนัช พนมเริงไชย

Packet Tracer - TCP and UDP Communications

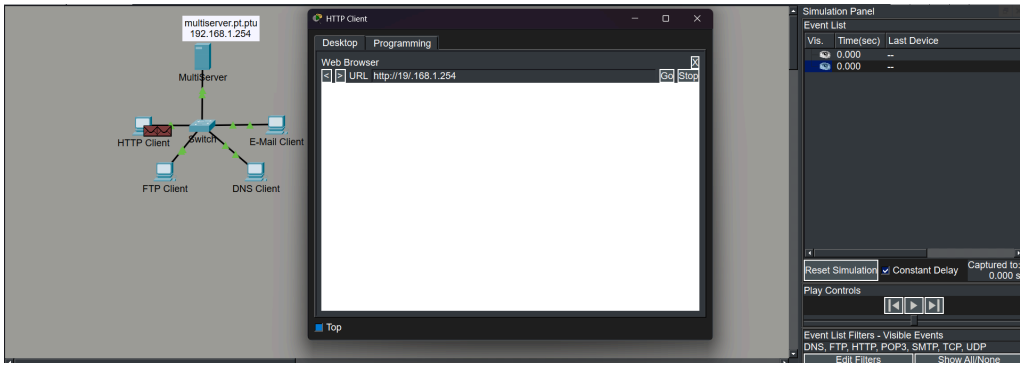
Instructions

Part 1: Generate Network Traffic in Simulation Mode and View Multiplexing

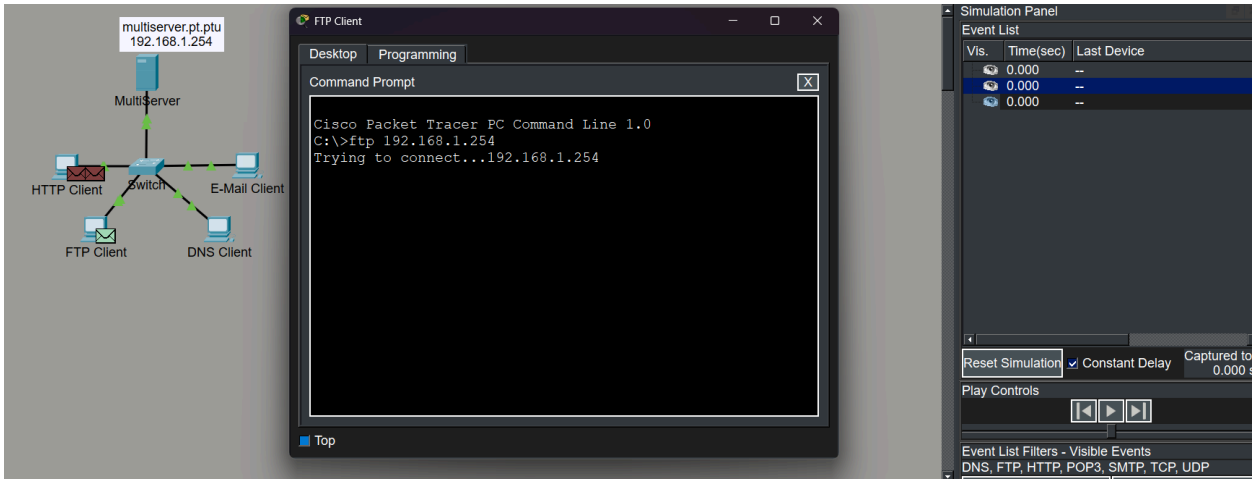
Step 1: Generate traffic to populate Address Resolution Protocol



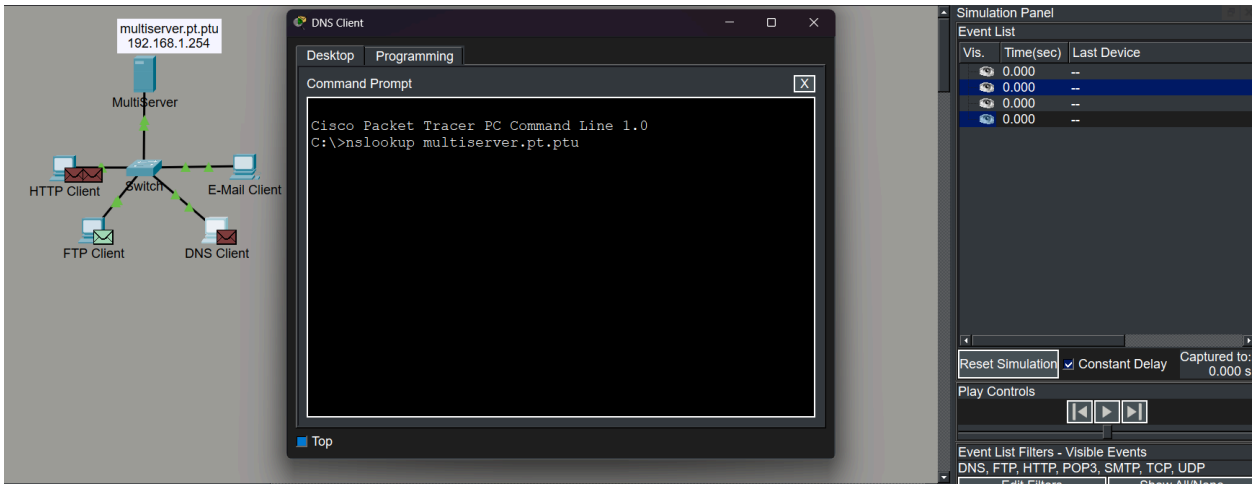
Step 2: Generate web (HTTP) traffic.



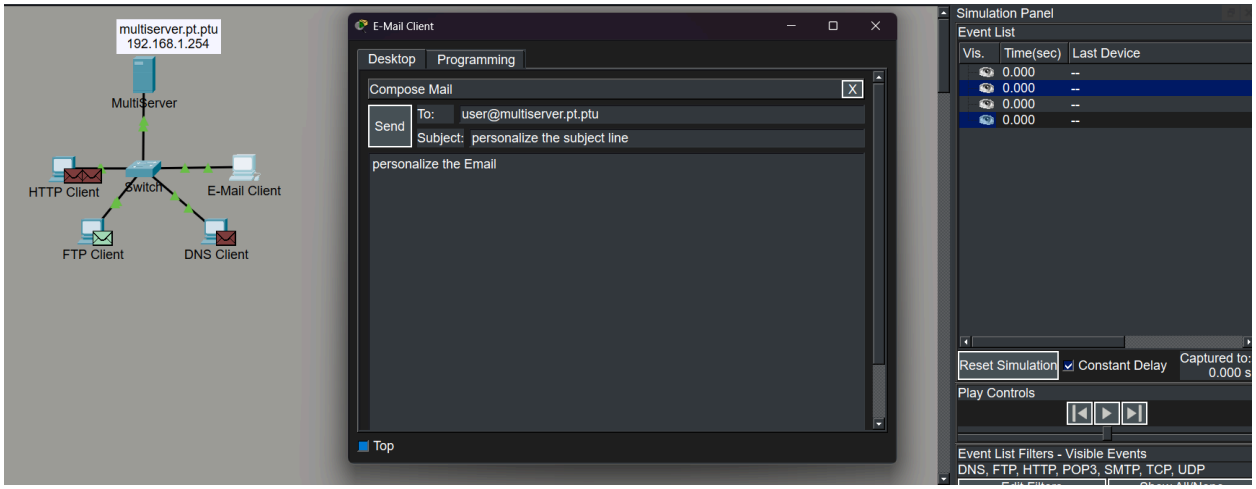
Step 3: Generate FTP traffic.

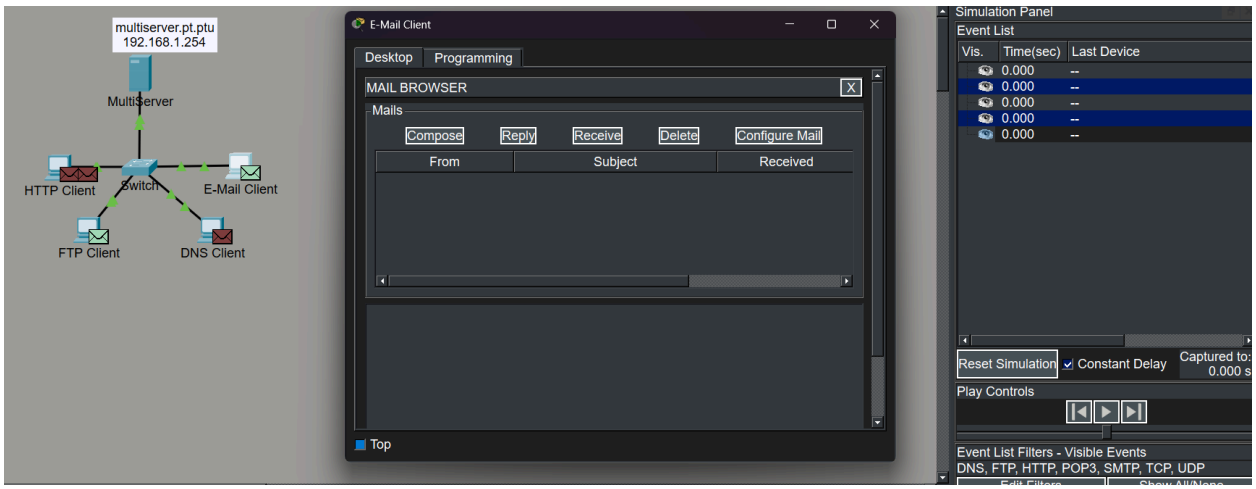


Step 4: Generate DNS traffic.



Step 5: Generate Email traffic.





Step 6: Verify that the traffic is generated and ready for simulation.

There should now be PDU entries in the simulation panel for each of the client computers.

Step 7: Examine multiplexing as the traffic crosses the network.

1 - What is this called?

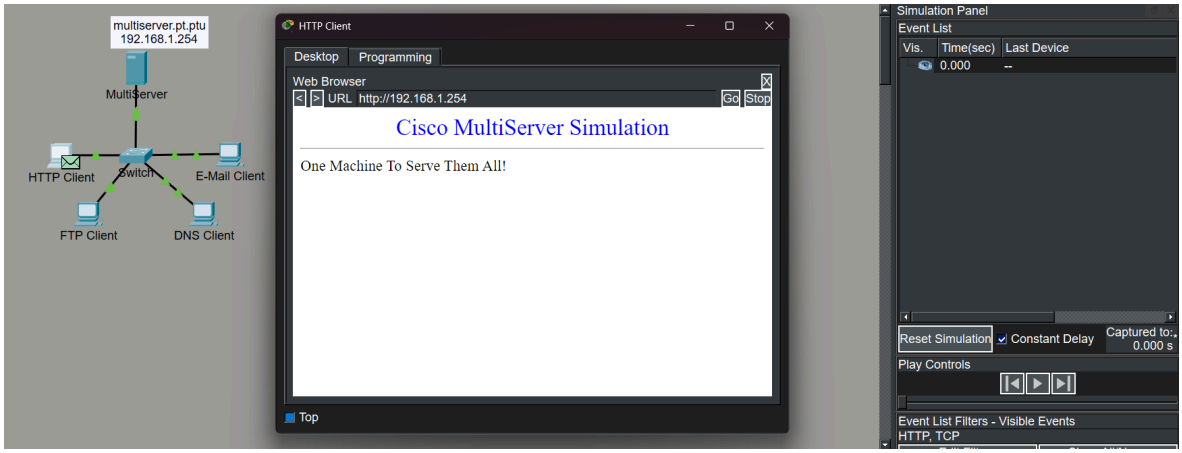
Ans. Multiplexing หลายโปรโตคอล ใช้สายสื่อสารเดียวกันได้ โดยสลับกันส่งข้อมูลทีละ PDU

2 - A variety of PDUs appears in the event list in the Simulation Panel. What is the meaning of the different colors?

Ans. สีแทนโปรโตคอลที่แตกต่างกัน HTTP, FTP, DNS, SMTP, POP3 ฯลฯ จะมีสีต่างกัน ใช้เพื่อให้แยกชนิดของ PDU ได้ง่ายใน Simulation Panel

Part 2: Examine Functionality of the TCP and UDP Protocols

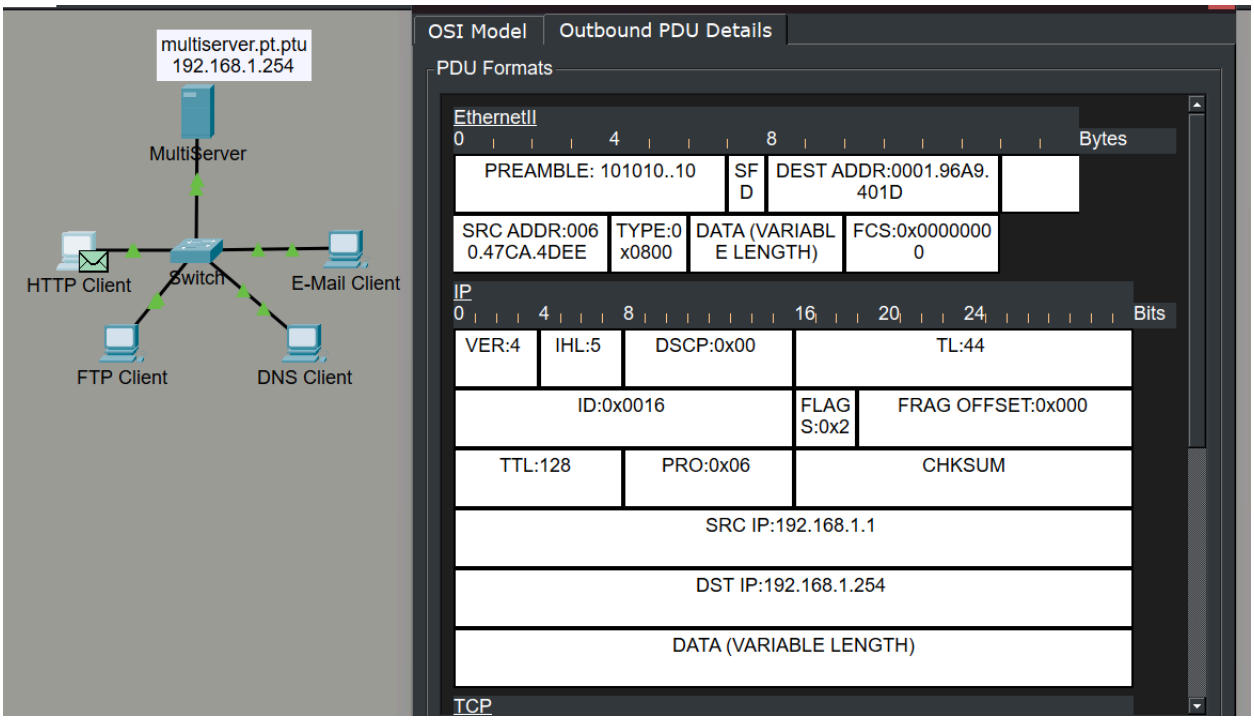
Step 1: Examine HTTP traffic as the clients communicate with the server.

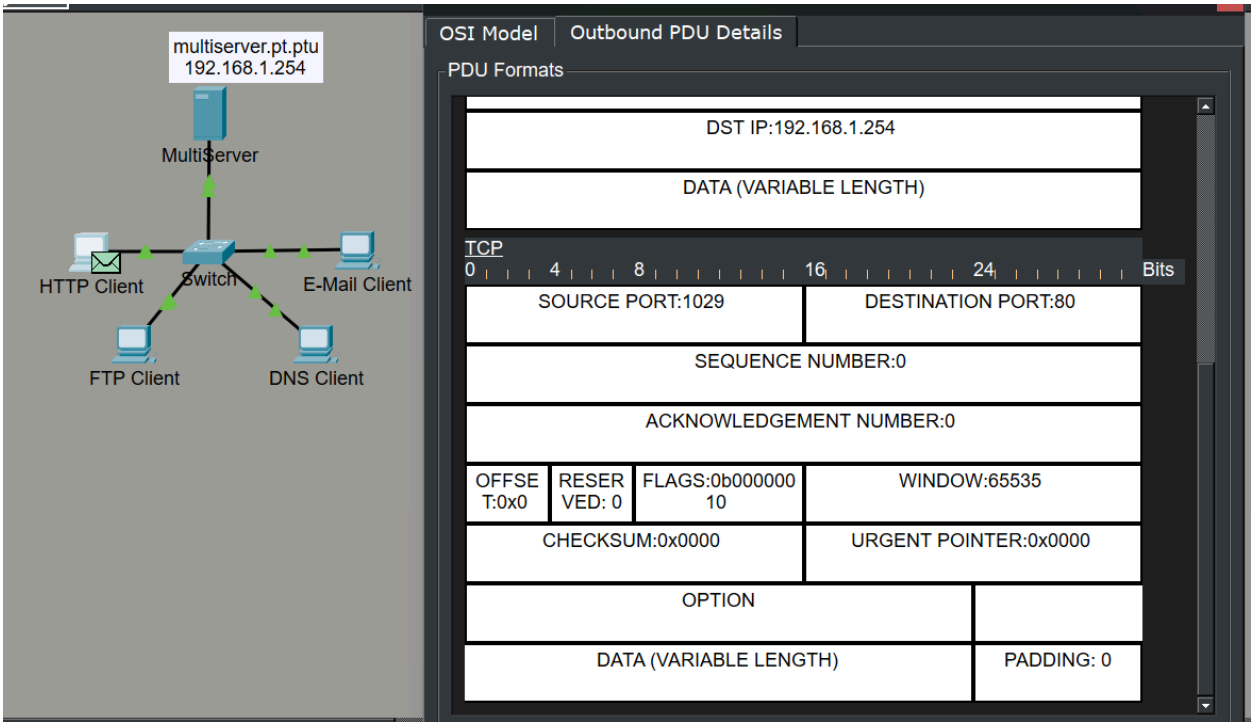


Why did it take so long for the HTTP PDU to appear?

Ans. เพราะ TCP ต้องทำ Three-Way Handshake (SYN → SYN-ACK → ACK) ก่อนเริ่มส่งข้อมูล HTTP จริง

e. Click the PDU envelope to show the PDU details. Click the **Outbound PDU Details** tab and scroll down to the second to the last section.





What is the section labeled?

Ans. **Transmission Control Protocol (TCP)**

Are these communications considered to be reliable?

Ans. ใช่ เพราะ HTTP ใช้ TCP

Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values.

Ans. **SRC PORT = 1029**

DEST PORT = 80

SEQUENCE NUM = 0

ACK NUM = 0

f. Look at the value in the Flags field, which is located next to the Window field. The values to the right of the "b" represent the TCP flags that are set for this stage of the data conversation. Each of the six places corresponds to a flag. The presence of a "1" in any place indicates that the flag is set. More than one flag can be set at a time. The values for the flags are shown below.

Flag Place	**6**	**5**	**4**	**3**	**2**	**1**
Value	URG	ACK	PSH	RST	SYN	FIN

Which TCP flags are set in this PDU?

Ans. SYN

g. Close the PDU and click **Capture/Forward** until a PDU with a checkmark returns to the **HTTP Client**.

h. Click the PDU envelope and select **Inbound PDU Details**.

How are the port and sequence numbers different than before?

Ans. SRC ↔ DEST สลับกัน, ACK เพิ่มขึ้น

i. Click the HTTP PDU which **HTTP Client** has prepared to send to **MultiServer**. This is the beginning of the HTTP communication. Click this second PDU envelope and select **Outbound PDU Details**.
What information is now listed in the TCP section? How are the port and sequence numbers different from the previous two PDUs?

Ans. มี HTTP data, Port เดิม Sequence / ACK เพิ่มขึ้นตามลำดับข้อมูล

j. Reset the simulation.

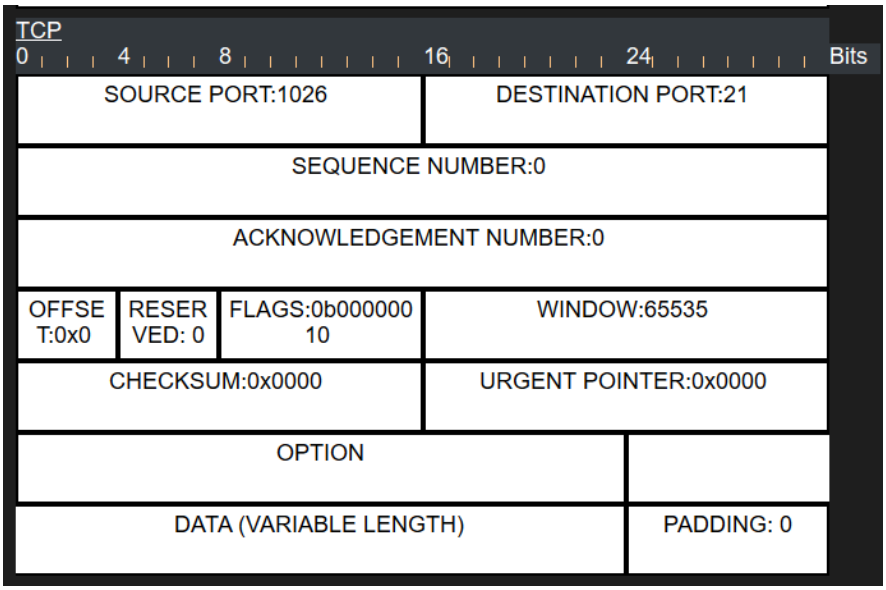
Step 2: Examine FTP traffic as the clients communicate with the server.

a. Open the command prompt on the FTP Client desktop. Initiate an FTP connection by entering **ftp 192.168.1.254**.

b. In the Simulation Panel, change **Edit Filters** to display only **FTP** and **TCP**.

c. Click **Capture/Forward**. Click the second PDU envelope to open it.

Click the **Outbound PDU Details** tab and scroll down to the TCP section.



Are these communications considered to be reliable?

Ans. ใช่เพราะ FTP ใช้ TCP

d. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values.
What is the value in the flag field?
Ans. 0b00000010 (SYN flag)

e. Close the PDU and click **Capture/Forward** until a PDU returns to the **FTP Client** with a checkmark.

f. Click the PDU envelope and select **Inbound PDU Details**.

How are the port and sequence numbers different than before?

Ans. พอร์ตต้นทางและปลายทางมีการสลับฝั่งกัน ส่วนค่า sequence number เริ่มที่ 0 และ acknowledgement number ยังเป็น 0 เนื่องจากเป็นช่วงเริ่มต้นของการเชื่อมต่อ TCP

g. Click the **Outbound PDU Details** tab.

How are the port and sequence numbers different from the previous results?

Ans. พอร์ตต้นทางและปลายทางไม่เปลี่ยนแปลงจากเดิม ส่วนค่า sequence number และ acknowledgement number ยังคงเป็น 0 เนื่องจากยังอยู่ในช่วงเริ่มต้นของการเชื่อมต่อ TCP

h. Close the PDU and click **Capture/Forward** until a second PDU returns to the **FTP Client**. The PDU is a different color.

i. Open the PDU and select **Inbound PDU Details**. Scroll down past the TCP section.
What is the message from the server?
Ans. Message:Welcome to PT Ftp server

j. Click Reset Simulation.

Step 3: Examine DNS traffic as the clients communicate with the server.

a. Repeat the steps in Part 1 to create DNS traffic.

b. In the Simulation Panel, change **Edit Filters** to display only **DNS** and **UDP**.

c. Click the PDU envelope to open it.

d. Look at the OSI Model details for the outbound PDU.

What is the Layer 4 protocol?

Ans. UDP

Are these communications considered to be reliable?

Ans. ไม่น่าเชื่อถือเนื่องจาก DNS ใช้ UDP ซึ่งไม่มีการรับประกันการส่งข้อมูล

e. Open the Outbound PDU Details tab and find the UDP section of the PDU formats. Record the **SRC PORT** and **DEST PORT** values.

Why are there no sequence and acknowledgement numbers?
Ans. เพราะ UDP เป็นโปรโตคอลแบบไม่เชื่อถือได้ (connectionless)
และ ไม่มีการใช้ sequence number หรือ acknowledgement number

f. Close the **PDU** and click **Capture/Forward** until a PDU with a check mark returns to the **DNS Client**.

g. Click the PDU envelope and select **Inbound PDU Details**.

How are the port and sequence numbers different than before?

Ans. Source port และ Destination port สลับฝั่งกัน (SRC ↔ DEST)
ยังไม่มี sequence number เพราะเป็น UDP

What is the last section of the **PDU** called? What is the IP address for the name **multiserver.pt.ptu**? ***Ans.*** Domain Name System (DNS)

h. Click Reset Simulation.

Step 4: Examine email traffic as the clients communicate with the server. a. Repeat the steps in Part 1 to send an email to **user@multiserver.pt.ptu**.

b. In the Simulation Panel, change **Edit Filters** to display only **POP3, SMTP** and **TCP**. c. Click the first PDU envelope to open it.

d. Click the **Outbound PDU Details** tab and scroll down to the last section.

What transport layer protocol does email traffic use?

Ans. TCP

Are these communications considered to be reliable?

Ans. น่าเชื่อถือเนื่องจากใช้ TCP

e. Record the **SRC PORT**, **DEST PORT**, **SEQUENCE NUM**, and **ACK NUM** values. What is the flag field value?

SRC PORT: พอร์ตสุ่ม (ephemeral port)

DEST PORT: SMTP → **25** POP3 → **110**

SEQUENCE NUM: 0

ACK NUM: 0

flag field value SYN (0b00000010)

f. Close the **PDU** and click **Capture/Forward** until a PDU returns to the **E-Mail Client** with a checkmark.

g. Click the TCP PDU envelope and select **Inbound PDU Details**.

How are the port and sequence numbers different than before?

Source port และ Destination port สลับฝั่งกัน
Sequence number เป็นของฝั่ง Server
Acknowledgement number เพิ่มขึ้นเพื่อยืนยันข้อมูล

h. Click the **Outbound PDU Details** tab.

How are the port and sequence numbers different from the previous two results?

Port ยังคงเดิม
Sequence number เพิ่มขึ้น
Acknowledgement number เปลี่ยนตามข้อมูลที่ได้รับ

i. There is a second **PDU** of a different color that **E-Mail Client** has prepared to send to **MultiServer**. This is the beginning of the email communication. Click this second PDU envelope and select **Outbound PDU Details**.

How are the port and sequence numbers different from the previous two **PDU**s?

Ans. Port ยังคงเดิมSequence number เพิ่มขึ้นอีกเป็นช่วงเริ่มต้นของ การส่งข้อมูลอีเมลจริง

What email protocol is associated with TCP port 25? ***Ans.*** SMTP

What protocol is associated with TCP port 110? ***Ans.*** POP3